Penetration Testing Report

Vstarcam IP Camera Security Assessment

March 4, 2026

Report Date	March 4, 2026
Tester	Hanna Hahn
Target System	Vstarcam IP Camera (x6 in lab environment)
Assessment Type	Vulnerability Assessment (Educational Lab)
Status	Complete

1. Executive Summary

This report documents a penetration testing exercise conducted on Vstarcam IP cameras in a controlled lab environment. The assessment revealed critical vulnerabilities including default credentials and exposed CGI endpoints that allow unauthenticated remote control of camera functions. An attacker with access to the network could remotely manipulate camera position, take snapshots, and disable surveillance without any authentication.

2. Lab Environment

2.1 Network Setup

- 6 Vstarcam IP cameras
- 6 switches
- 1 Fortinet firewall (central gateway)
- Multiple ethernet cables
- 12 students (2 per subnet)

2.2 Objective

Identify and exploit vulnerabilities in the Vstarcam cameras to gain remote access and execute commands (specifically to take a snapshot photo of ourselves to subimt).

3. Reconnaissance and Discovery

3.1 Initial Network Scanning

Step 1: Ping Sweep

- `nmap -sn <firewall_ip>`
- Result: did not reveal individual subnets that we wanted.

Step 2: Subnet Range Scanning

- `nmap -sn 192.168.1-225.1`
- Result: Discovered 7 IP addresses including the firewall and each subnet gateway.

3.2 Port Discovery

Step 3: Initial Port Scan

- `nmap -sV <IP>`
- Result: Only 2 ports open, not enough information.

Step 4: Full Port Scan

- `nmap -sV -p- <IP>`
- Result: Identified camera IP addresses and service ports.

4. Vulnerability Discovery

4.1 Finding 1: Default Credentials

Severity	**High**
Vulnerability Type	Default Credentials / Weak Authentication
CVSS Score	8.8
Affected Component	Web Admin Panel
Status	Confirmed

4.1.1 Description

The Vstarcam camera ships with hardcoded default credentials. Rather than attempt brute force, I researched the camera model online and discovered the manufacturer's default username and password.

4.1.2 Technical Details

- URL: `http://<camera_IP>:<port>`
- Authentication: HTTP form-based login
- Tools: Burp Suite for traffic analysis.

4.2 Finding 2: Exposed CGI Endpoints

Severity	**High**
Vulnerability Type	Unprotected API Endpoints / Broken Access Control
CVSS Score	8.2
Affected Component	CGI control interface
Status	Confirmed

4.2.1 Technical Details

Vulnerable CGI Endpoint: `/cgi-bin/api.cgi`

Example Commands:

- Pan control: `http://<IP>:<PORT>/decoder_control.cgi?&command=<cmd>&onestep=0`
- Snapshot: `http://<IP>:<PORT>/snapshot.cgi?user=*****&pwd=******`

5. Remediation Recommendations

1. Change default credentials immediately upon camera deployment.
2. Implement authentication and authorization checks on all CGI endpoints.
3. Restrict CGI API access to authenticated users only.
4. Use HTTPS/TLS for all camera communications.
5. Implement network segmentation to restrict camera access.

6. Conclusion

The Vstarcam cameras in this lab environment contain multiple critical vulnerabilities that allow unauthorized remote access and control. These vulnerabilities are easily exploitable and should be remediated immediately in any production environment.

Appendix: Evidence and Screenshots

Lab Network Setup

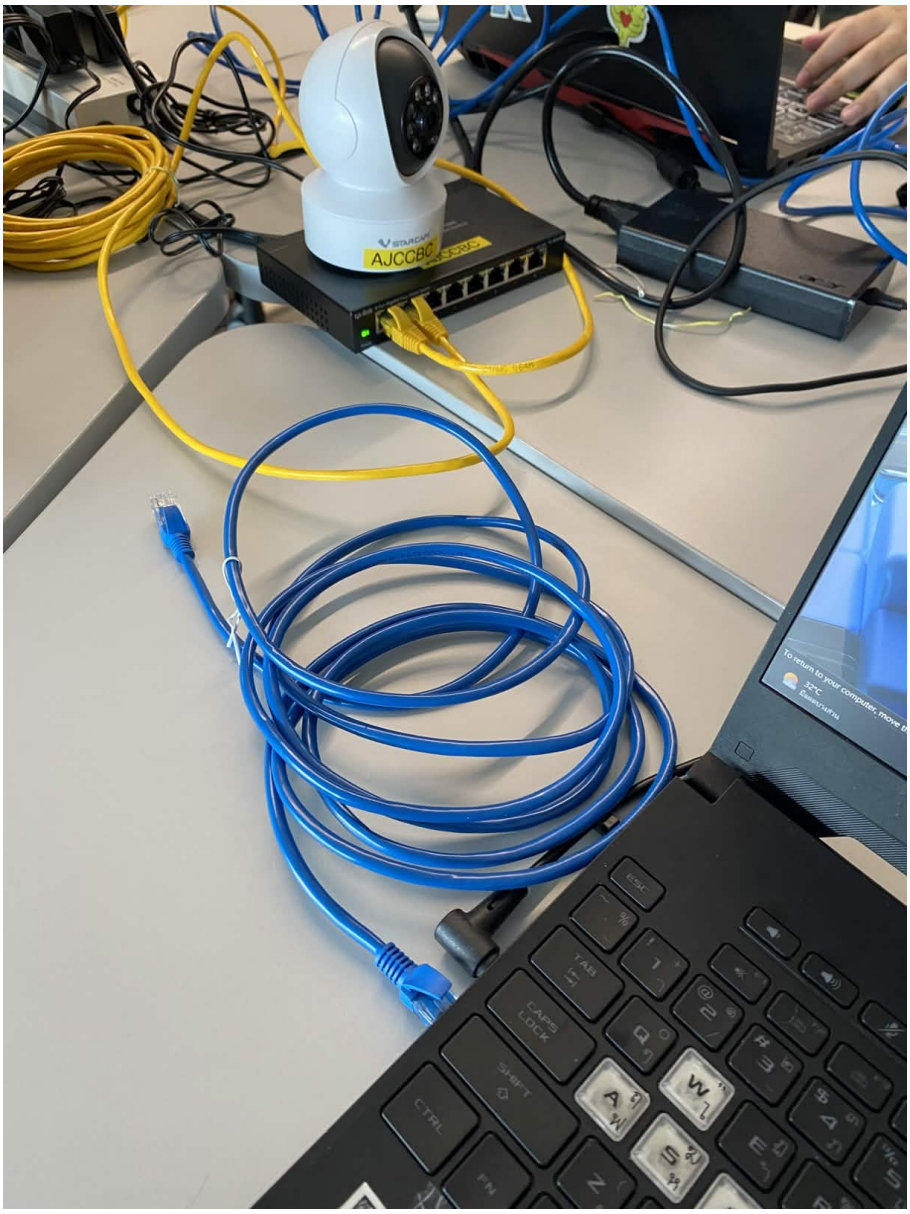

Figure 1: Lab environment showing Vstarcam IP camera connected to network switch with yellow and blue ethernet cables. The white camera is mounted on the black network switch, demonstrating the physical setup used for this assessment.

Live Camera Feed Exploitation

Figure 2: Real-time camera feed displayed on laptop screen. This demonstrates successful access to the camera's live video stream after exploiting the CGI endpoint vulnerability. The image shows multiple students in the lab captured through the Vstarcam camera.

Proof of Concept: Snapshot Capture

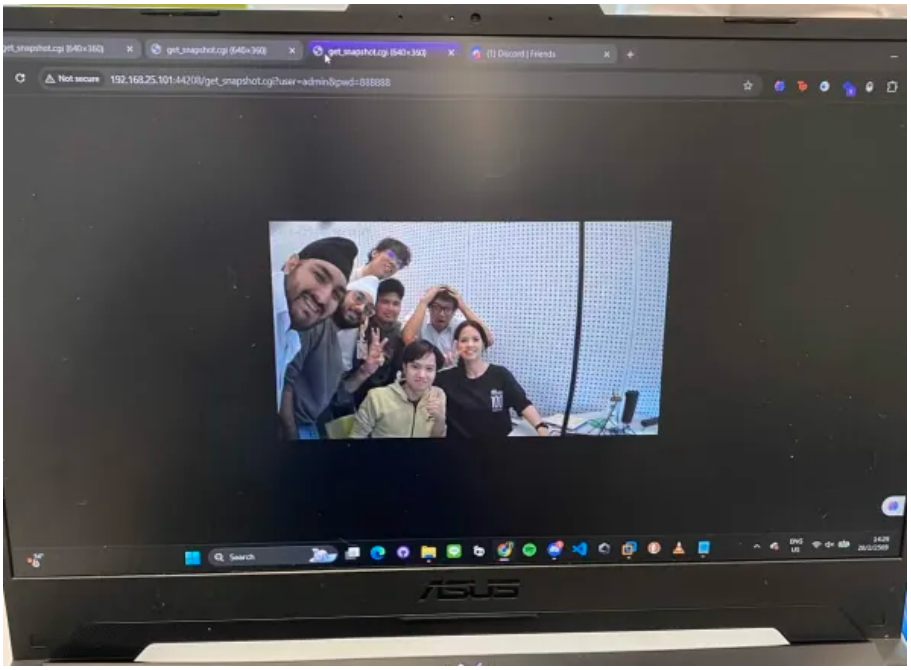

Figure 3: Successful snapshot photograph taken using the CGI command exploit. This image shows the Vstarcam camera positioned to capture a view of the lab environment, demonstrating complete control over camera functions through the identified vulnerabilities.